A Black-Box Function Optimization Framework

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Model overview

- **Model name:** Imperial Capstone Project Model
- **Task:** identify a promising point for evaluating a target function in search of an optimum, when the value of the function is only known for a limited (small) number of points in the domain.
- **Type:** The model consists of two parts: (i) an SVM framework to identify regions of the domain which seem promising for a next candidate; (ii) a Bayesian optimization framework, which allows a precise determination of the optimal next point on the basis of a trade-off between exploring promising new regions of the domain and exploiting areas near already-known candidates.
- **Deployment:** the model can be used in any setting where function evaluation is expensive, and can only be done in a limited way, so grid searches are impractical.

Intended use

- **Primary tasks:** identifying optimal values of a target function.
- **Target users:** researchers and practitioners who face constraints (technical, resources, or time, or otherwise) in terms of how many times they can explore different combinations of values for input variables in search of the optimal one.
- **Use case:** making constrained optimal decisions under limitations regarding function evaluation or experimentation.
- **Limitations:** the framework cannot be guaranteed to reach an overall global optimum, so care should be taken in using it, as well as understanding the tradeoffs involved in model decisions. For instance, greater emphasis on exploration may limit the framework's ability to explore well-known regions and to reach global optima; greater emphasis on exploitation may leave many unexplored regions, and lead the framework to miss promising areas altogether.

Training data

- **Size:** the model contains a hyperparameter (kappa) which trades off decisions between exploration and exploitation. The choice of this value was based on a simulation which used a fixed experiment length and dimension for the domain. The code for this simulation is available in this repository. Inasmuch as real-world situations vary, users may wish to perform their own simulations to set this parameter in an appropriate way.

- **Sources:** both SVM and Bayesian Optimization are standard techniques in Machine Learning. The calibration of how kappa evolves over the experiment was based on hypothetical functions which have single and multiple optima.
- **Features:** The model was initially applied to the Capstone project in Imperial College's Certificate Program in AI/ML. The features vary according to the experiment, but, inasmuch as this is an optimization-based approach, the model can in principle be applied to any setting which can be reduced to evaluating a target function of multi-dimensional real inputs.

Inputs and outputs

- **Input:** for each function, previous values of variables and function values.
- **Output:** the next suggested data point for function evaluation.

Ethical considerations

- **Bias risks:** the model has mainly been used to determine optimal function values for a variety of types of datasets. Any bias risks would need to be identified in the context of each setting in which the model is applied.
- **Transparency:** the data for which the model was first used is not publicly available. However, initial data points for any hypothetical target function can be easily generated, and these can be used to train the model.
- **Recommendations:** as function evaluation becomes cheaper and easier, the question of whether the model is appropriate needs to be revisited.

Distribution

- Model card and code available in this repository.